

Which Comparison a Fixed Effect Draws

TEACHING NOTE

ACCTG 597A • Week 7

On Kropko and Kubinec (2020), “Interpretation and Identification of Within-Unit and Cross-Sectional Variation in Panel Data Models,” *PLOS ONE* 15(4): e0231349. Running example: Anantharaman and Lee (2014, *JFE*).

A&L’s two tables give two answers

- At A&L’s evaluation point, the CFO vega slope falls from 14.8 to 2.6 when firm effects replace industry effects.

Pension equity share, Model 3 (× 1,000)	Table 5: industry and year	Table 6A: firm and year
CFO vega slope at the 95th percentile of distress (computed here)	14.8	2.6
Distress × CFO vega, A&L’s test of H2	2.050 (t = 2.02)	1.010 (t = 1.38)

Printed vega coefficients (slopes at distress = 0): 18.6 and 4.5.

Bet for the room: which column measures the effect of CFO vega.

“A correct answer to the wrong question is as useless as the wrong answer to the right question.”

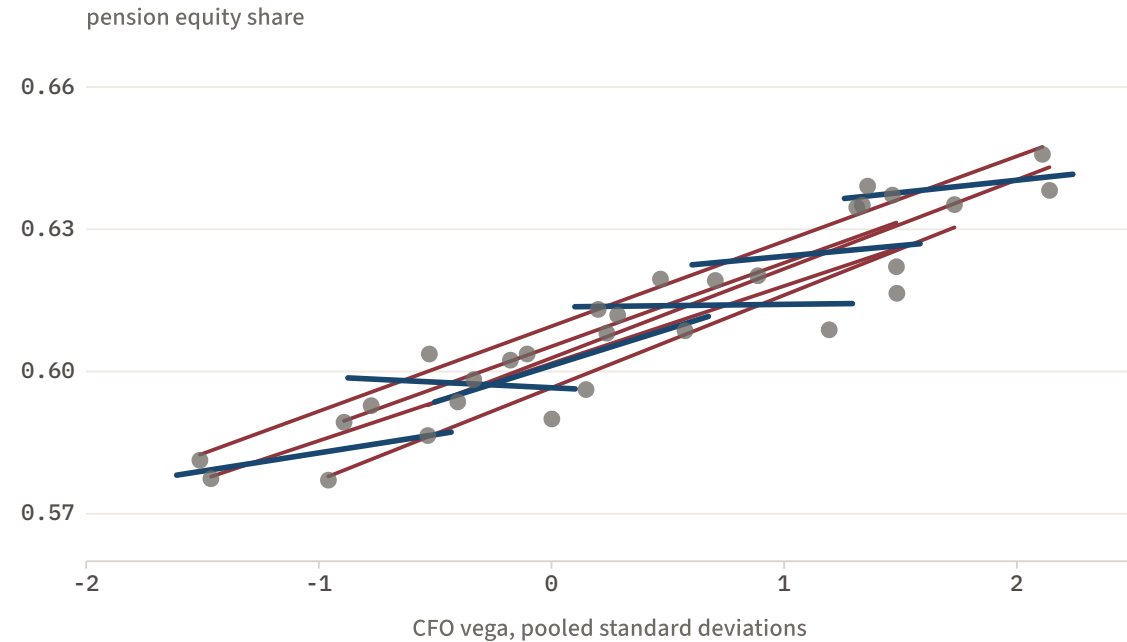
Kropko and Kubinec (2020, p. 2)

A panel can be sliced by firm or by year

- The same firm-years support two comparisons: one firm over time, or many firms within one year.

Maroon lines compare firms within a year; navy lines follow one firm over time. A one-way fixed effect averages one family.

Illustrative draw from the simulated panel (seed 2014): six firms, five years.



Across firms, each year

Within each firm, over time

Maroon: years, average 17.9. Navy: firms, average 5.5 (per 1,000)

Systematic part of the outcome, with the residual noise scaled down so that the lines are visible.

Each one-way slope is a weighted average

Firm effects: K&K Eqs. (10)–(11)

$$\hat{\beta}_{\text{firm FE}} = \frac{\sum_i \omega_i \hat{\beta}_i}{\sum_i \omega_i}, \quad \omega_i = \sum_t (x_{it} - \bar{x}_i)^2$$

Year effects: K&K Eq. (8) and p. 8

$$\hat{\beta}_{\text{year FE}} = \frac{\sum_t \omega_t \hat{\beta}_t}{\sum_t \omega_t}, \quad \omega_t = \sum_i (x_{it} - \bar{x}_t)^2$$

$\hat{\beta}_i$: firm i 's time-series slope; $\hat{\beta}_t$: year t 's cross-sectional slope.

- Firm effects average firms' time-series slopes; year effects average years' cross-sectional slopes.

K&K reserve firm effects for research that “intends to compare one case to itself over time” (p. 8).

Both hold to 0.0000000000 in the simulation. Room: Fama-MacBeth.

Two-way effects compare deviations each year

K&K Eq. (12), balanced panel

$$\hat{\beta}_{TW} = \frac{\sum_i \sum_t (y_{it} - \bar{y}_i - \bar{y}_t + \bar{y})(x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})}{\sum_i \sum_t (x_{it} - \bar{x}_i - \bar{x}_t + \bar{x})^2}$$

K&K Eq. (13): firm-demean, then one regression per year

$$y_{it}^* = \alpha_t + \beta_t^* x_{it}^* + \varepsilon_{it}^*, \quad x_{it}^* = x_{it} - \bar{x}_i$$

The two agree

$$\hat{\beta}_{TW} = \frac{\sum_t \omega_t^* \hat{\beta}_t^*}{\sum_t \omega_t^*}, \quad \omega_t^* = \sum_i (x_{it}^* - \bar{x}_t^*)^2$$

- Two-way effects first remove each firm's mean, then compare firms within each year.

K&K: two-way effects are “a generalization of the effect of deviations from the case-means at a particular point in time” (p. 20).

All three agree to 0.0000000000 in the simulation; exact only in a balanced panel.

Each choice keeps different slices of vega

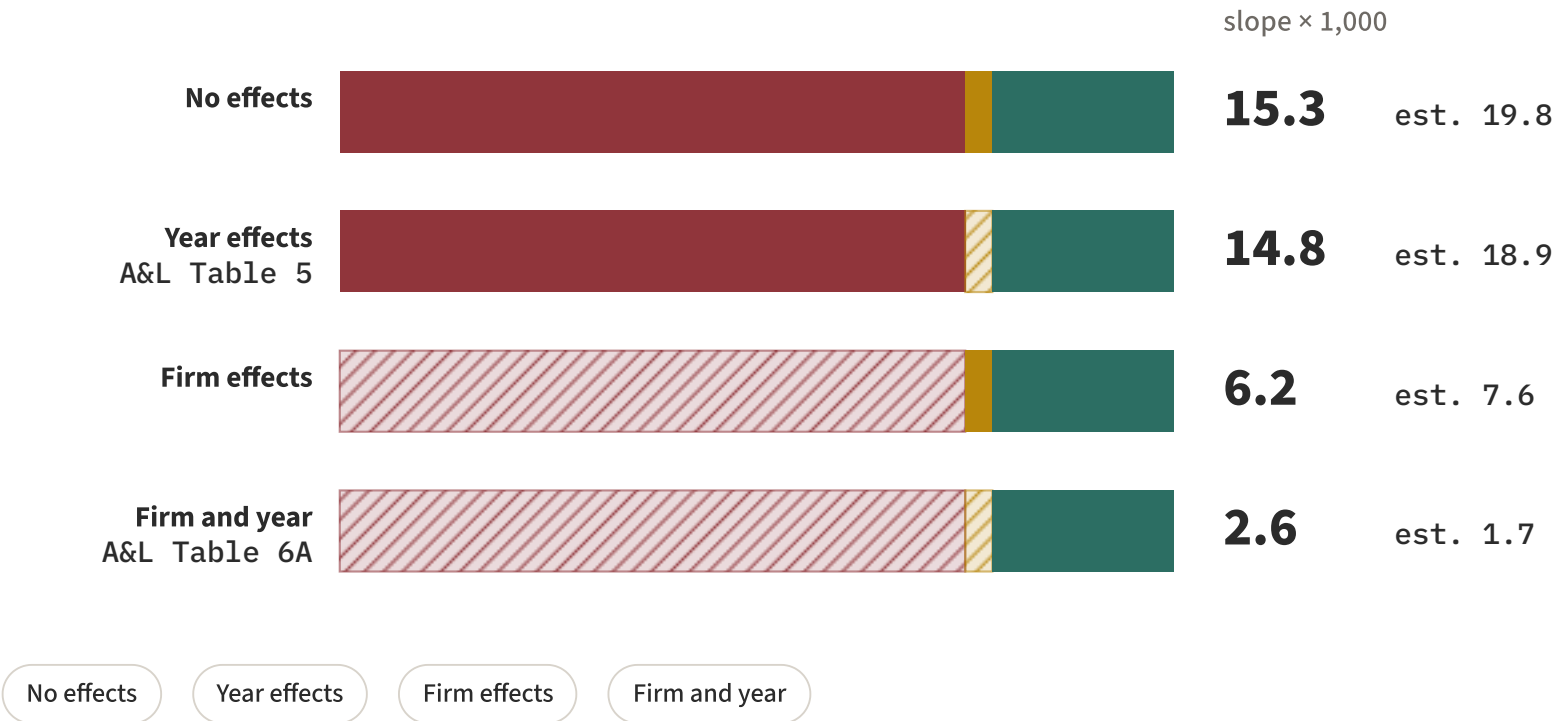
- Each choice keeps some slices of vega’s variance and returns the weighted average of their slopes.

Vega splits exactly into between-firm, year, and residual parts.

K&K grant that fixed effects remove time-invariant confounders, but “it is not correct to say that FE models control for confounding variables” (p. 9).

Decomposition: this note’s. Assumed: 25 percent of vega’s variation within firms; slopes set to match A&L at the 95th percentile of distress.

■ between firms: 75.0% of vega's sum of squares, slope 18.3 ■ common to a year: 3.3% of vega's sum of squares, slope 30.0
■ neither: 21.7% of vega's sum of squares, slope 2.6



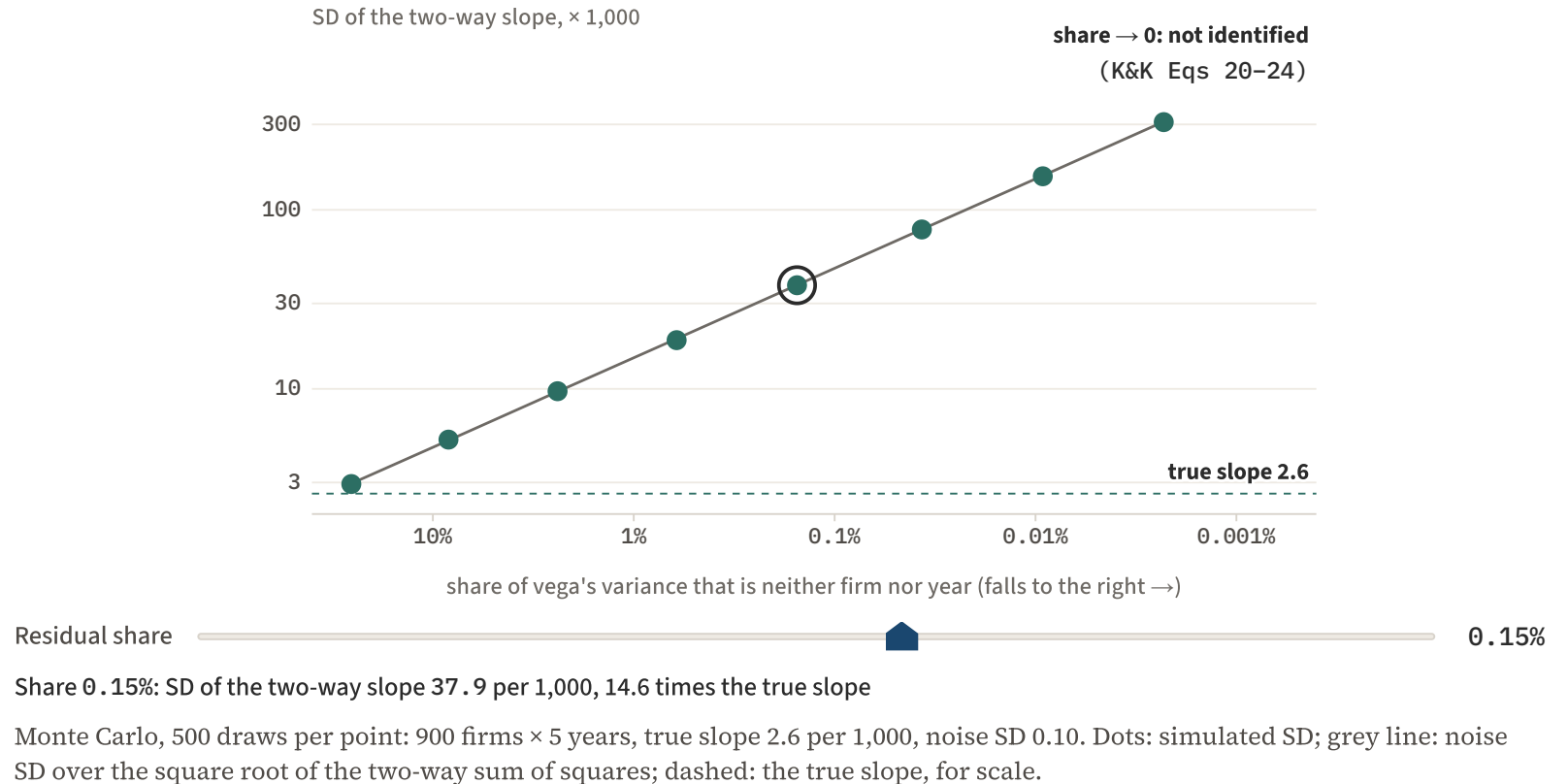
Simulated panel (seed 2014). Hatched slices are removed by the fixed effects. Bold: slope on the systematic part, the share-weighted average of the kept slices' slopes; "est.": estimate in one simulated draw.

Additive vega leaves the slope unidentified

- If vega were exactly a firm part plus a year part, two-way demeaning would leave nothing.

In K&K's design, homogeneous (and unequal) within-year and within-firm slopes make x additive (Eqs. 20–24), so “the two-way FE specification is statistically unidentified” (p. 1). With firm-year variation of its own, x identifies the slope, less precisely as that variation shrinks.

Rooms: [what Stata reports](#) · [K&K's design](#).



A&L’s two tables answer two questions

■ H2 is stated across firms; Table 5 compares firms without removing persistent traits, Table 6A makes a narrower comparison.

	A&L Table 5: industry and year	A&L Table 6A: firm and year
Comparison	high- versus low-vega CFOs’ firms, net of industry means and year means	a firm’s deviation from its own mean vega, net of common year shifts
H2: “Risk shifting through pension asset allocation to equities is stronger in firms whose managers have high vega” (p. 332)	the comparison H2 states	a within-firm question net of common shocks
Keeps	persistent firm traits that travel with vega	only firm-year variation in vega

A&L present firm effects as a check that “alleviate to some degree the concerns of omitted, time-invariant, firm- or manager-specific factors” (p. 337). An omitted trait and a weaker within-firm relation produce the same pattern.

State the comparison, then the fixed effects

■ Choose the fixed effects that isolate the stated comparison, and report what each choice keeps.

The hypothesis states	Fixed effects	Keep, and report
Firms with high x differ in y, same year	year (and industry)	persistent firm traits: name them
A firm’s y moves when its x moves	firm	common shocks
The same, net of common shocks	firm and year	only the residual part: report its share

Firm and year effects remove persistent traits and common shocks together; the cost is a narrower comparison.

“Fixed effects models address omitted variable bias by accounting for time-fixed covariates, case-fixed covariates, or both. But they also change the research question being evaluated by the model.”

Kropko and Kubinec (2020, p. 20)

Room: firm versus two-way slopes.

Fixed effects choose the comparison

- 1** One-way fixed effects average subset slopes: firm effects answer an over-time question, year effects a cross-sectional one, and the answers can differ.
- 2** Firm and year effects use only the regressor's residual part; when it is small the estimate is imprecise, and when it is absent the slope is not identified.
- 3** The comparison the hypothesis states should choose the fixed effects; they are not a robustness check.

Kropko and Kubinec (2020) · Anantharaman and Lee (2014) · github.com/sbb151/acctg597a-research-design

Appendix

Firm and two-way slopes differ by a year term

Frisch-Waugh-Lovell in a balanced panel

$$\hat{\beta}_{\text{firm FE}} - \hat{\beta}_{TW} = \frac{N \sum_t \hat{d}_t (\bar{x}_t - \bar{x})}{\sum_{i,t} (x_{it} - \bar{x}_i)^2}$$

$\hat{d}_t$: the year effects estimated in the two-way model.

- Adding year effects moves the firm slope only through the year means of x.

The gap depends on the year component's share of within-firm variation (13 percent here) and on how the year effects covary with it, not on the number of firms.

One simulated draw: firm effects 7.60, firm and year 1.70 ($\times 1,000$); the identity holds to 0.0000000000 (check 5). A derivation for this note, not stated by K&K. [Back](#)

Stata drops a dummy and returns a K&K slope

- When x is exactly additive, Stata drops one more dummy, and the slope it reports depends on the order of the dummies.

In K&K's homogeneous design (within-year slope 1, within-firm slope -3 , Eq. 21), `regress y x i.case i.time` drops 30.time and returns **-3.000** ; `regress y x i.time i.case` drops 30.case and returns **1.000** .

“Generally speaking, if the dropped FE happens to be a dummy variable for a case, then the resulting coefficient on x resembles the time FE coefficient on x .”

Kropko and Kubinec (2020, p. 18)

30 × 30 panel, noise-free outcome ([kk_components.do](#)). [Back](#)

K&K's design gives x no shock of its own

- In K&K's simulations each observation is where a firm line meets a year line, so x has no shock of its own.

K&K posit $E(y_{it}) = \alpha_t + \beta_t x_{it}$ and $E(y_{it}) = \alpha_i + \gamma_i x_{it}$ at once (Eqs. 14–16), which forces $x_{it} = (\alpha_i - \alpha_t) / (\beta_t - \gamma_i)$ (Eq. 17); the only non-additive variation in x comes from slope heterogeneity. Their Simulation 1 (Fig. 6, p. 15) shows two-way effects landing between the two mean slopes, with a larger variance.

With firm-year variation of its own in x , two-way effects recover a common slope, and individual estimates in K&K's design can fall outside the interval between the mean slopes.

A reading developed for this note. [Back](#)

Fama-MacBeth weights years equally

- Year effects and Fama-MacBeth average the same annual cross-sectional slopes with different weights.

A regression with year effects only (K&K Eq. 8) equals the p. 8 weighted average of annual slopes, with weights $\omega_t = \sum_i (x_{it} - \bar{x}_t)^2$; Fama and MacBeth average the same $\hat{\beta}_t$ with weights $1/T$. Both answer the cross-sectional question; they differ in weights and in how their standard errors treat dependence across years.

An observation for this note, not a claim in K&K. [Back](#)